# Standard Data Products

APPN Aerial SOP — Locked revision 1.0

## Standard Data Products

This page provides an overview of the **standard data products** generated by APPN UAV processing pipelines, including typical spatial resolution, file formats, and software compatibility. By default, products are delivered in **open or widely supported formats** that can be accessed using **freely available tools**.

For the pipelines that generate these products, see Processing Pipelines.

For additional infomation about all all GOBI and CALVIS products see the Gryfn Documentation

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### Guiding principles

- Products are generated using standardised, documented pipelines to ensure consistency across platforms and sites.
- Output formats prioritise accessibility using freely available and open-source software.
- Spatial resolution and product definitions are explicit and traceable.
- Products are intended to be suitable for downstream analysis, comparison, and reuse.

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### Standard LiDAR-derived products

| Product | Description | Typical resolution | Typical file size *(indicative)* |
|---|---|---|---|
| LiDAR Digital Surface Model (DSM) | Surface model representing canopy and above-ground features | 8 cm | Derived from LiDAR point c |
| LiDAR Digital Terrain Model (DTM) | Ground surface model with vegetation removed | 1 m | Derived from LiDAR point c |
| Combined LiDAR Point Cloud | Aggregated and filtered point cloud | Native point spacing | ~110–175 MB per minute of f |

*Typical LiDAR file sizes are based on raw LiDAR logging rates documented in the CALViS and GOBI SOPs; DSM and DTM sizes scale with survey area and output resolution.*

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### Standard hyperspectral products

| Product | Description | Typical resolution | Typical file size *(indicative)* | Form |
|---|---|---|---|---|
| VNIR Orthomosaic | Radiometrically calibrated VNIR hyperspectral mosaic | ~4 cm (GSD-based) | Several GB per flight (even for short missions) | ENV |
| SWIR Orthomosaic | Radiometrically calibrated SWIR hyperspectral mosaic | ~4 cm (GSD-based) | Several GB per flight (even for short missions) | ENV |

*Hyperspectral data volumes scale strongly with mission duration, frame period, and number of spectral bands. SOPs note that even short flights routinely generate multi-GB datasets.*

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## Standard RGB products

| Product | Description | Typical resolution | Typical file size *(indicative)* | Format |
|---|---|---|---|---|
| RGB Orthomosaic | High-resolution RGB orthorectified mosaic | ~0.6 cm (fixed) | ~30–70 MB per image (image every ~2 s by default) | GeoTIFF (`.tif` |